

DIN-2025-0094

A DATA-DRIVEN MODELING STRATEGY FOR FLUID-STRUCTURE INTERACTION PROBLEMS FROM TIME-DELAYED STRUCTURAL INFORMATION

Lucas Franceschini

franceschinilc@gmail.com

Paolo Furlanetto Ferrari

Guilherme Franzini

Celso Pupo Pesce

Escola Politécnica da Universidade de São Paulo - Offshore Mechanics Laboratory

José Roberto Castilho Piqueira

Escola Politécnica da Universidade de São Paulo - Systems Engineering

Abstract. *This work deals with a data-driven methodology for modeling the Vortex-Induced Vibration (VIV) phenomenon. In particular, we aim at reconstructing, purely from data, an autonomous dynamical system representing the time-evolution solely of the structural variables. The theoretical justification behind this pursuit lies ultimately on Takens' embedding theorem stating that a given dynamical system (describing, in our case, the VIV phenomenon where both solid and fluid are coupled together) is conjugate to another system acting on the time-delay coordinates of a given measure (in our case the solid displacement only), from which we could compute its numerical time-derivatives. We show that, for the case-study of a rigid cylinder in a two-dimensional low-Reynolds flow, for a range of reduced velocities (from pre to post lock-in conditions), the full dynamics can be well characterized by three delayed coordinates, regarded as the position, velocity and acceleration of the structure's displacement, as a regular mechanical system would. We further show that this dynamics can be reconstructed by a regression procedure, minimizing the error between a nonlinear polynomial model and the data. This model recovers the relevant features of the original system in the entire span of reduced velocities we considered, providing a minimal reduced-order model preserving the most relevant features of the input signal.*

Keywords: *Fluid-Structure Interaction, Vortex-Induced Vibration, data-driven modeling, time-delayed embedding*

1. INTRODUCTION

Fluid-Structure Interaction (FSI) are highly complex problems that arise in many context in engineering (such as aeronautical and offshore industries). Because of this complexity, the study of those systems, oftentimes require cumbersome and costly numerical simulations, requiring a Computational Fluid Dynamics (CFD) solver, coupled with a structural one (see Hou *et al.* (2012)). Both CFD and structural solvers are in themselves vast and complex active fields of research (see Anderson and Wendt (1995); Bhatti *et al.* (2020); Bittnar and Sejnoha (1996)). Furthermore the multiphysics coupling also represents a vast and active research field (see Hou *et al.* (2012)). Alternatively, FSI experiments may be an adequate tool for studying such systems, its main drawback being the balance between the cost of the experiment and the amount of data acquired (and its precision). Indeed, oftentimes only solid-related quantities can be measured (for example, by solid-mounted load-sensors, laser-positioning sensors (see Franzini *et al.* (2015)) and/or video cameras (see Defensor Filho and Pesce (2023); Madi *et al.* (2019))) when fluid-related quantities (such as the velocity field) are only available through more sophisticated methods, such as Particle Image Velocimetry (PIV, see Adrian and Westerweel (2011)).

This inaccessibility of the fluid variables and the high complexity of numerical simulations motivate us to exploit the solid data at best. With this in mind, we propose in this work a methodology for the identification of an autonomous dynamical system, written solely for the solid variables, and its time-derivatives. The ultimate theoretical justification for this approach resides in the work of Takens (see Takens (2006); Sauer *et al.* (1991)), which states that any (smooth) dynamical system, acting on a given state-space, can be (diffeomorphically) mapped onto the time-delayed coordinates of a (also smooth) scalar measurement of the state. The number of delayed-coordinates to be considered is a delicate question, as it must be sufficiently high for us to be able to write a dynamical system on those new coordinates, but not too large, to avoid having to deal with numerous coordinates, which can be problematic for the next step, which will be the writing of the surrogate (meta-model) for the system's dynamics by regression (due to over-fitting issues and/or by computational reasons, see for example Bhatti *et al.* (2020); Kaiser *et al.* (2018)). Indeed, in this regression step, the data (on those time-delayed coordinates) will be used to "train" a nonlinear (meta- or surrogate) model, based on polynomial nonlinearities. We remark that the dimension of such system is much lower than the original one, allowing us to have an

easy-to-simulate phenomenological model replacing demanding CFD simulations and/or costly experimental campaigns.

This procedure will be applied to a Vortex-Induced Vibration (VIV) problem, where a rigid cylinder is mounted on a spring and subjected to a laminar two-dimensional flow. We point out this is an academic case, which represents a simplification with respect to many other theoretical, experimental and numerical results, for example taking into account higher-Reynolds numbers flows (Sarpkaya (1979); Bearman (1984)), flexibility of the cylinder (Pesce and Fajfarra (2000)), three-dimensional effects of the cylinder and the flow (Fajfarra *et al.* (2001); Fajfarra (2002); Franzini *et al.* (2009)). However, despite its simplicity, we consider this is a good test case for our methodology.

2. METHODOLOGY

First of all, we consider phenomena that can be entirely described by a state-vector $\mathbf{x} \in \mathbf{R}^n$, where n is the dimension. We will also suppose that the time-evolution of this vector is given by the following general nonlinear system:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) \quad (1)$$

Alternatively, we may consider the flow associated with this system, that advances the state-vector by a finite time-lag τ as:

$$\mathbf{x}(t + \tau) \equiv S\mathbf{x}(t) = \mathbf{x}(t) + \int_t^{t+\tau} \mathbf{f}(\mathbf{x}(s))ds \quad (2)$$

This system is often too complex to be handled in an efficient manner, as it often requires costly numerical simulations and/or experimental campaigns. Indeed, in our cases, this would represent a highly-dimensional and highly nonlinear Fluid-Structure Interaction phenomenon. This motivates us to create surrogate/phenomenological models uniquely for some quantity of interest, in our case, the solid displacement/velocity/acceleration variables. To do so, we will rely on time-delayed coordinates, as discussed in the next section.

2.1 Time-delayed of measurement as coordinates

To the state-vector $\mathbf{x}$, we perform a scalar measurement:

$$y \equiv h(\mathbf{x}) \quad (3)$$

We time-delay the measurements:

$$\Phi(\mathbf{x}) = (h(\mathbf{x}), h(S\mathbf{x}), \dots, h(S^{d-1}\mathbf{x})) = (y_0, y_1, \dots, y_{d-1}) \equiv \mathbf{y} \quad (4)$$

Takens (see Takens (2006); Sauer *et al.* (1991)) showed that this transformation $\Phi(\mathbf{x})$ is a diffeomorphism (namely it is a continuous differentiable onto function, having an inverse on its image, also continuous differentiable) for *any* sufficiently smooth measurement $h(\mathbf{x})$ provided that the dimension of the time-delayed coordinates d is high enough. Typically, if $d \geq 2n + 1$ ensures the diffeomorphical character of $\Phi(\mathbf{x})$. However, in practice, a much lower number of delayed coordinates is sufficient. Indeed, if we desire to only capture the attractor, of (possibly fractal) dimension n_a , the amount of necessary delayed-coordinates is $d \geq 2n_a + 1$.

If we define $\mathbf{y}_n = \mathbf{y}(t_n) = \mathbf{y}(t_0 + n\tau)$, we can write a system taking the previous time delayed-vector $\mathbf{y}_n$ to the next one $\mathbf{y}_{n+1}$ as:

$$\mathbf{y}_{n+1} = \Phi \circ S\mathbf{x}_n = \Phi \circ S \circ \Phi^{-1}(\mathbf{y}_n) = \tilde{\Phi}(\mathbf{y}_n) \quad (5)$$

Or, by recognizing that for the first $d - 1$ entries of the vector $\mathbf{y}_{n+1}$ the system can be trivially given by $(\mathbf{y}_{n+1})_j = (\mathbf{y}_n)_{j+1}$ (for $j < d$), the whole dynamics can be cast for the last entry of $\mathbf{y}_{n+1}$ as $y_{n+d} = g(y_n, y_{n+1}, \dots, y_{n+d-1})$ or simply:

$$y_{n+1} = g(y_n, y_{n-1}, \dots, y_{n-d+1}) \quad (6)$$

Before continuing, we remark that the fact that we only perform a scalar operation on $\mathbf{x}$, $h(\mathbf{x})$ is, by no means, a restrictive feature, as this is the "most-difficult" scenario, where only one scalar information is available, and more non-redundant measurements can be added, possibly decreasing this way the delay dimension d .

2.2 Time-derivatives of measurements as coordinates

The theoretical results presented in the previous sections were meant to justify building a dynamical system for the delayed coordinates only. However, instead of trying to model the dynamics for y_{n+1} given the previous measurements, we will try to provide a functional relation for the numerical derivatives, defined recursively with the aid of a second-order central finite difference scheme as:

$$\frac{d^k y}{dt^k}(t_n) \approx [D^k y]_n = \frac{[D^{k-1} y]_{n+1} - [D^{k-1} y]_{n-1}}{2\tau} \quad \text{with} \quad [D^0 y]_n = y_n \quad (7)$$

This is more interesting as the structure of the system will be much less reliant on the actual time-delay size (τ), and will provide a much more readily physically interpretable dynamics. An equivalent form of the dynamical system (6) will be then recast it as:

$$g(D^{d-1} y_n, D^{d-2} y_n, \dots, y_n) = 0 \quad (8)$$

where here an implicit relation for, say, $D^{d-1} y_n$ is sought. With some abuse of notation, for the sake of simplicity, we will confound the numerical derivatives' notation with then continuous ones $D^k y \equiv d^k y / dt^k$, leading to:

$$g\left(\frac{d^{d-1} y}{dt^{d-1}}, \dots, y\right) = 0 \quad (9)$$

2.3 Reconstruction of nonlinear dynamics through regression of data

This section is devoted to the construction of an approximation for the unknown dynamics $g\left(\frac{d^{d-1} y}{dt^{d-1}}, \dots, y\right)$. Here, instead of proposing a general model for it "from scratch", we will try to incorporate, as much as we can, some prior knowledge of the system, given by g_{prior} and leave to the data-driven regression a complement δg as:

$$g = g_{prior} + \delta g, \quad \text{with} \quad \delta g \approx \sum_{i_0}^{N_0} \dots \sum_{i_{d-1}}^{N_{d-1}} \gamma_{i_0, \dots, i_{d-1}} y^{i_0} \dot{y}^{i_1} \dots \left(\frac{d^{d-1} y}{dt^{d-1}}\right)^{i_{d-1}} \quad (10)$$

This unknown (data-driven) part, as shown, will be written as a polynomial in each of its arguments. The coefficients $\gamma_{i_0, \dots, i_{d-1}}$ of each monomial will be found by performing a least-square regression, where we minimize the cost functional:

$$J(\gamma_{i_0, \dots, i_{d-1}}) = \frac{1}{2} \left\| g_{prior} - \sum_{i_0}^{N_0} \dots \sum_{i_{d-1}}^{N_{d-1}} \gamma_{i_0, \dots, i_{d-1}} y^{i_0} \dot{y}^{i_1} \dots \left(\frac{d^{d-1} y}{dt^{d-1}}\right)^{i_{d-1}} \right\|_2^2 + \frac{\sigma}{2} \sum_{i_0}^{N_0} \dots \sum_{i_{d-1}}^{N_{d-1}} \gamma_{i_0, \dots, i_{d-1}}^2 \quad (11)$$

being the second term a regularization component, penalizing equally the square of each coefficient $\gamma_{i_0, \dots, i_{d-1}}$. Although different weights could be attributed according to the different nature of each term, for now, this simple form will be considered, which is already good enough to prevent overfitting the data, as we shall see.

2.4 Remarks on the choice of the model

Before going further to the actual physical configuration we will treat, we briefly discuss on the choice of the model δg . In the form we presented, we let it be an arbitrary polynomial function on every coordinate, $d^k y / dt^k$. However, letting it be nonlinear polynomials on the last time-derivative $d^{d-1} y / dt^{d-1}$ can be problematic for the usefulness of the model, as it would require an implicit solver to determine this quantity in function of the other lower-order derivatives (which not only is cumbersome numerically, but may not have a solution or even have more than one). For this reason, we will restrict ourselves to linear approximations on that quantity, namely $N_{d-1} = 1$.

3. Application to Fluid-Structure Interaction

3.1 A Vortex Induced Vibration (VIV) problem

We consider the two-dimensional fluid structure interaction problem consisting on a rigid cylinder, with mass m , mounted on a spring of stiffness k , acted on by a steady current. The crosswise displacement of the cylinder, Y , is given by:

$$m \frac{d^2}{dT^2} Y + kY = L(\mathbf{U}, P) = \int_{\Gamma_s} \mathbf{e}_y \cdot (P \cdot \mathbf{I} - \mu(\nabla_{\mathbf{x}} \mathbf{U} + \nabla_{\mathbf{x}} \mathbf{U}^T)) \mathbf{n} dS \quad (12)$$

where $L(\mathbf{U}, P)$ is the lift force, computed as an integral over the solid wall Γ_s of an incompressible fluid-flow related variables, namely the velocity field $\mathbf{U} = \mathbf{U}(\mathbf{X}, T)$, the pressure field $P = P(\mathbf{X}, T)$, the density ρ and viscosity μ . The fluid motion is modeled by the Navier-Stokes equations:

$$\rho(\partial_T \mathbf{U} + (\mathbf{U} - \mathbf{W}) \cdot \nabla_{\mathbf{x}} \mathbf{U}) + \nabla_{\mathbf{x}} P = \mu \Delta_{\mathbf{x}} \mathbf{U}, \quad \nabla_{\mathbf{x}} \cdot \mathbf{U} = 0 \quad (13)$$

Here we make a choice of writing the Navier-Stokes equations in the Arbitrary Lagrangian Eulerian (ALE) form (see for example Hirt *et al.* (1997)), as it will be the choice for our numerical implementation when taking the mesh-velocity, $\mathbf{W}$, as the cross-wise velocity of the cylinder, $\mathbf{W} = \mathbf{e}_y \dot{Y}$. If we use the incoming flow velocity U_∞ , the cylinder diameter D and its mass m_c as dimensional parameters, we can express time, space, cylinder displacement, flow velocity and pressure in dimensionless for as:

$$t = T(U_\infty/D), \quad \mathbf{x} = \mathbf{X}/D, \quad y = Y/D, \quad \mathbf{u} = \mathbf{U}/U_\infty, \quad p = P/(\rho U_\infty^2) \quad (14)$$

which can be used to rewrite the equations also in a non-dimensional form as:

$$\partial_t \mathbf{u} + (\mathbf{u} - \mathbf{w}) \cdot \nabla_{\mathbf{x}} \mathbf{u} + \nabla_{\mathbf{x}} p = Re^{-1} \Delta_{\mathbf{x}} \mathbf{u}, \quad \nabla_{\mathbf{x}} \cdot \mathbf{u} = 0$$

$$\ddot{y} + \left(\frac{2\pi}{U_r} \right)^2 y = \left(\frac{2}{\pi m^*} \right) C_L \quad (15)$$

where $Re = \rho U_\infty D / \mu$ is the Reynolds number (taken equal to $Re = 100$ in all our cases, as in Menon and Mittal (2021)), $C_L = 2L/(\rho D U_\infty^2)$ is the (instantaneous) lift coefficient, $U_r = U_\infty / D f_n$ is the reduced velocity, given by the ratio of the fluid pass-through-solid time and the solid natural period in air $T_n = 1/F_n = 2\pi\sqrt{m/k}$ (to be varied in our experiments) and $m^* = m/(\rho\pi D^2/4)$ the reduced mass, given by the ratio of the cylinder's mass and the fluid-displaced mass (taken as $m^* = 10 \cdot (2/\pi) \approx 6.4$, as in Menon and Mittal (2021)). We remark that by taking both the Reynolds and the reduced mass constants, we fix both the vortex-shedding frequency (characterized by a Strouhal number of $St = F_{VS} D / U_\infty \approx 0.18$) and the mass/diameter of the cylinder. For this reason, if we were to reproduce those settings experimentally, we would have to vary the stiffness of the spring on which the cylinder is mounted (instead of increasing the flow velocity, which is more usual in practice). Although less common, this setup could still be realized, as it was done in Franzini *et al.* (2008). Furthermore, if we were to take the frequency of $y(t)$, as given in function of the dimensionless time, it would be proportional to the dimensional one as $F = (D/U_\infty)f$.

In order to study this problem, we simulate it using the finite-element code FreeFEM++ (see Hecht (2012)) for the spatial discretization of the fluid problem, a second-order backward finite difference for the time-derivative of the fluid problem (see Karniadakis *et al.* (1991)) and a centered finite-difference in time is used for the solid equations.

The results of several time-stepping simulations, started with initial conditions corresponding to the steady-state (base-flow) solution added with the linear mode (unstable, corresponding to the vortex-shedding mode, and multiplied by a small amplitude) can be appreciated in Figure 1 for $U_r = 4, 4.5, 5$ and 8 . We can see that we recover the usual behavior where for small reduced velocities, the amplitude of motion is small and it is caused by the forcing from the fluid, which oscillates at the vortex-shedding (VS) frequency, typically characterized by a Strouhal number $St = F_{VS} D / U_\infty \approx 0.18$. For $U_r \geq 5$, the solid natural frequency matches the VS one, and a resonance occurs, increasing the amplitude of motion. This is the beginning of the lock-in region. If we further increase U_r , we still witness a large amplitude of motion and the frequency of that motion is locked to the solid's natural frequency, even if the VS one is not matching exactly that frequency. We see that the term $F/F_n = f \cdot U_r = 1$ becomes unity in this region. For much larger reduced velocities, near $U_r = 8$, we begin to witness the end of this phenomenon, where the amplitude of the motion becomes again smaller and the frequencies in the signal slowly start to transition to the fluid one. A summary of this description is given in Figure 2 showing the amplitude of motion and the Power Spectral Density (PSD) of the signal (always computed for $t > 100$, the fully-converged portion) for each reduced velocity considered here. We remark that, although this results are very close from the one reported by Menon and Mittal (2021), it is qualitatively different from a more usual lock-in diagram where, not only higher amplitudes are obtained, but also a lower branch after it seems to be missing (see for example, Fujarra *et al.* (2001)). This is due to the simplifications in our model, for the sake of the methodology presentation and validation.

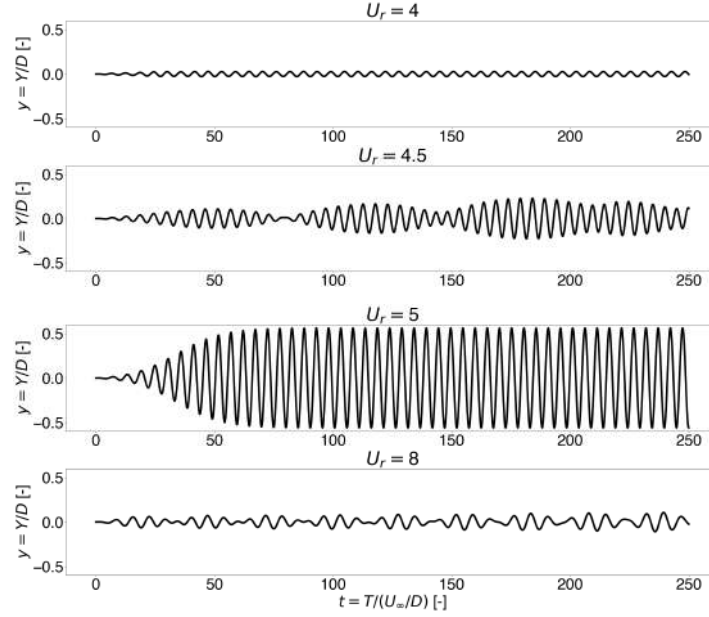


Figure 1. Signal from cross-wise (y) of the cylinder for $U_r = 4, 4.5, 5$ and 8 , determined from CFD simulations.

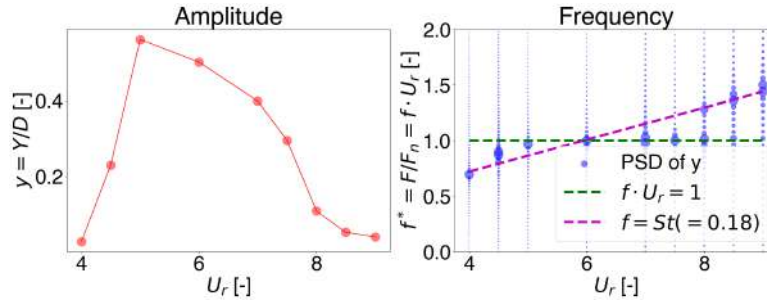


Figure 2. Amplitude and Power Spectral Density (PSD) for displacement signal $y(t)$ for several reduced velocities. The typical lock-in scenario is recovered where high-amplitudes are obtained for a reduced-velocity interval starting from $U_r \approx 5$, from which its frequency (computed from the dimensionless time t) stops being correlated with the vortex-shedding one (characterized by a Strouhal number of $St \approx 0.18$) and correlates with the natural frequency of the solid $f_n = U_r$ (also measured dimensionless) the solid.

3.2 Motivating a second-order model for the VIV

Before presenting the regression results, it is important to comment on the number of time-delayed coordinates to be considered or, alternatively, the number of derivatives in our model ($d - 1$), together with the actual dynamical mode we will consider for the regression. To try to address this issue, we analyze the trajectories $(y, \dot{y}, \ddot{y})$ as a "copy" of our state-space, for the fully developed range only ($t > 100$). Those trajectories are shown in Figure 3, together with (in colors) the actual forcing term coming from the fluid flow, $C_L(t)$.

In those plots, we can see that for $U_r = 4$ (pre lock-in) and $U_r = 5$ (lock-in), the trajectories are (almost nearly) periodic, which can be modeled as a second-order mass-spring-like system. For this reason, the projection of the trajectories on the plane $(y, \ddot{y})$ lies on a straight line. We point out that this straight line does not, in general, have the "inclination" from the solid system in air (in other words its natural frequency), as the actual system is forced by the fluid. For the pre-lock-in case ($U_r = 4$), this force is periodic and animates the solid at the same frequency, coming from the vortex-shedding (VS) phenomenon. Since this frequency is not the one from the solid, a model that would fit this data could be given by an added-mass one, as we shall see in the next sections. For the lock-in case ($U_r = 5$), the frequency becomes the one from the solid, and no such correction would be needed. We observe that, for those reduced-velocities, we can be hopeful that a close-form system can be found for $(y, \dot{y}, \ddot{y})$. In other words, this means that no extra derivatives is needed for the incorporation of the fluid forces (exogenous to the solid) into a modified second-order model for the solid only.

For the $U_r = 4.5$ and $U_r = 8$, transition regions between a Locked-In states and a fluid-dominated one, we have a more complex scenario. For $U_r = 4.5$, the trajectory on the space $(y, \dot{y}, \ddot{y})$ is no longer a simple curve closed on itself anymore, showing that it is no longer periodic. Indeed, the signal not only has a single frequency, but also a modulation (already seen in Figure 1), which widens the trajectories forming a disk-like structure in the space $(y, \dot{y}, \ddot{y})$. Typically, such modulation

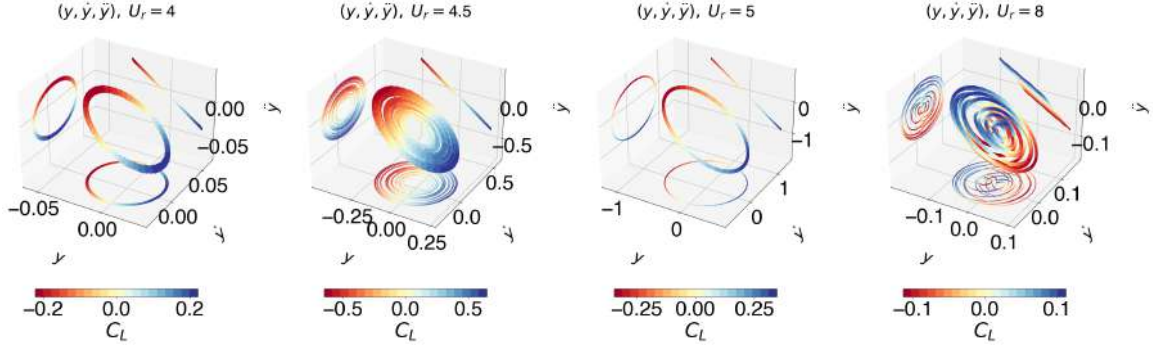


Figure 3. Three-dimensional plot of consecutive delayed coordinates, rewritten in the form of the displacement, velocity and acceleration signals, $(y, \dot{y}, \ddot{y})$ for $U_r = 4, 4.5, 5$ and 8 . We can see the bcorrelation between the acceleration $\ddot{y}$ and position and velocity $(y, \dot{y})$, suggesting there should be an underlying functional relation between them.

may appear when the signal contains two very close frequencies (the acoustic beat phenomenon) whereby the resulting signal will contain an amplitude (modulation) oscillating with half the difference of those two (very close) frequencies, multiplied with another signal oscillating with their average. This scenario seems to be the case as in this region both solid and vortex-shedding frequencies are very similar but not quite the same yet. Since the modulation varies slowly, and the actual oscillation can be identified with a single frequency, the projection of the trajectory onto the $(y, \dot{y})$ plane seems to be a line still, suggesting that a linear mass-spring model could be a good fit for the reconstructed dynamics. However, this linear dynamics alone is unable to reproduce the modulation, where, for it to happens an exogenous force would have to come into play, rendering the system non-autonomous. For this reason, we can predict that a model regression on those coordinates will perform sub optimally. This observation could imply that: either (a) the regression model would have to be very complex, containing several nonlinear terms and exploiting the (very small) width of $(y, \dot{y})$ projection or (b) more delayed-coordinates would be necessary to properly model the case $U_r = 4.5$. The hypothesis (a) could be effectively employed, for that particular choice of U_r . However, for slightly higher U_r , where the two frequencies would be even closer, the complexity of the model could be arbitrarily vast. Indeed, for that case, an infinitely thin line would be obtained in the $y, \dot{y}$ plane, making the trajectories $(y, \dot{y}, \ddot{y})$ to cross each other, whilst not forming a periodic orbit, a clear indication that more coordinates should be added for us to have an autonomous nonlinear model.

Let us consider now the case $U_r = 8$. This is yet another transitional case, but one for which two rather distinct frequencies coexist (see Figure 2). For this reason, the system no longer behaves as a single frequency, modulated by a slow one as it was the case for $U_r = 4.5$. Indeed, the projection of the trajectory on the $(y, \dot{y})$ plane is no longer a thin line, it is a little thicker. For this reason, we could be more hopeful to obtain a working-model than the lower order dynamics (than for the case $U_r = 4.5$) case.

3.3 Regression Results

In this section, we will show the results for the model-regression, based on the regularized least-square problem discussed previously in section 2.3 First of all, we shall comment on the prior model, which should contain the "well-understood" part of the dynamics, g_{prior} . As the we know the motion of the solid in air is modeled by a linear spring-mass oscillator, the following model will be considered:

$$g_{prior} = \ddot{y} + \left(\frac{2\pi}{U_r} \right)^2 y \quad (16)$$

For this reason, the model δg will be responsible for providing the force of the fluid $\left(\frac{2}{\pi m^*} \right) C_L(t)$, which was already given (as the colors) in Figure 3. We can see that, despite of the already pointed out issues at $U_r = 4.5$ and $U_r = 8$, the forcing $C_L(t)$ seems to be reasonably well correlated with $(y, \dot{y}, \ddot{y})$, suggesting that some functional relation should be obtained.

Let us consider now the forcing C_L , at the trajectories projected on the planes $(y, \dot{y})$. As pointed out before, this projection is either a straight line ($U_r = 4, 5$ and possibly 4.5) or a very thin region of that space ($U_r = 8$). For the former case, we can see that the forcing signal changes drastically its value in very small regions of the $(y, \dot{y})$ plane, which could imply in a model with several nonlinear terms. In the limit where this think region becomes a line ($U_r = 4, 5$ and 4.5), the issue is aggravated as an infinite number of models δg could be found. Indeed, let us take the case $U_r = 4$, where the "inclination" of that line has to be changed, in order to adjust to the fluid frequency. This change can be either done by varying the coefficient multiplying $\ddot{y}$ or y . The first one is the most usual choice, as it translates to the added-mass concept (see Cunha *et al.* (2006); Fajarra (2002)) and the second, less common, associated to a correction of the stiffness.

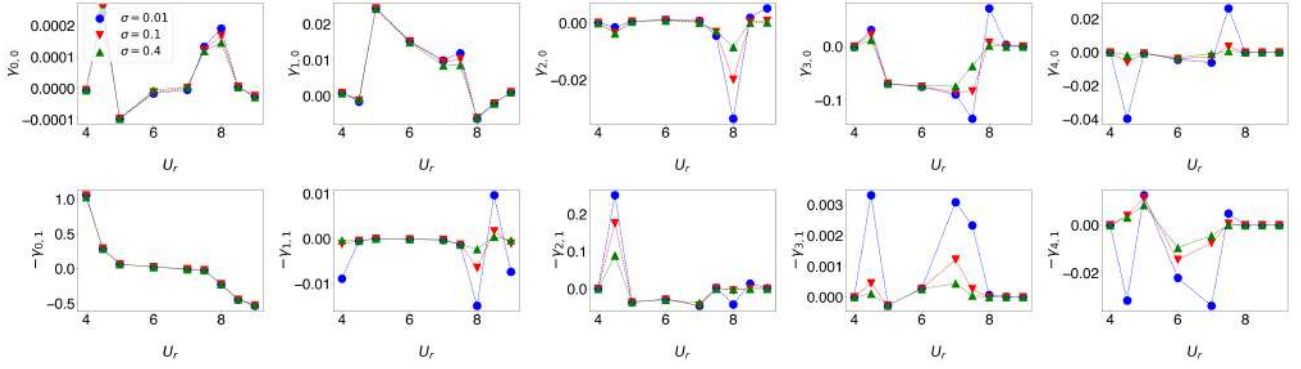


Figure 4. Regression results for model's coefficients $\gamma_{i,j}$ (where $i = 0, \dots, N_1$ representing the coefficients for the velocity dependency $\dot{y}$ and $j = 0, 1$ the acceleration $\ddot{y}$).

For this reason, we will only take $\ddot{y}$. We point out however that keeping y could be advantageous in some cases, when an explicit spatial dependence of the forcing exists. As an example, we could think of fluid flow geometries presenting solid walls, density/pressure gradients or upstream wakes (see, for example Assi *et al.* (2013)). Also, we believe that an explicit spatial dependence could be useful when the solid's velocity is much higher than the moving velocity of the fluid's vortices, experiencing them in a "quasi-steady" manner.

Let us consider now the projections of the trajectories on the plane $(\dot{y}, \ddot{y})$. We can see that, in that case, the trajectories project to circles or circle-like structures and, for all cases, each point on that plane has only (roughly) color, corresponding to C_L . This shows that this is a good candidate of variables to represent δg . It is interesting to notice that this is not the case for $(y, \dot{y})$ as in the case $U_r = 8$ the values for C_L exhibit different values for a same $(y, \dot{y})$ (two different color coinciding on the same region).

For those reasons, we shall consider a model for δg without any dependency on y ($N_0 = 0$), linear dependency on $\ddot{y}$ and a polynomial dependency on $\dot{y}$. In Figure 4 we present the regression results for $N_1 = 5$ and for three different values for the regularization parameter $\sigma = 0.01, 0.1$ and 0.4 . We can see that four parameters are robustly obtained for the three values of σ , which are $\gamma_{0,0}$, $-\gamma_{0,1}$, $\gamma_{1,0}$. The first one corresponds to a constant-value in the model, and is quite small, as the time-average of the original signal is also very small, and thus could be neglected. The second one $-\gamma_{0,1}$ coincides with the added-mass coefficient (with a negative sign, since it appears on the left-hand-side, with the mass). This behavior of the coefficient, which has a positive value before the lock-in, a negligible value within the lock-in and a negative value post lock-in is a classical result, reported for example in Fujarra (2002); Cunha *et al.* (2006). The third and fourth, $\gamma_{1,0}$ and $\gamma_{3,0}$ physically represents a (small) linear and cubic damping coefficients. The remaining terms oscillate with the variation of σ . For high values of σ , the regularization becomes more important than the benefit their presence would cause in the regression procedure, and they all become negligible. For lower values of σ , they tend to be important, especially for $U_r = 4.5$ and $U_r \approx 8$. Interestingly, those two cases were already discussed in previous sections and noted to be problematic, demanding rather complicated models to well capture the dynamics.

In Figure 5, we present results where we animate the model, by a time-stepping procedure, and compare with original signals. As expected, for $U_r = 4$ and $U_r = 5$, the model, for the three regularization parameters, perform very well. For the case $U_r = 4.5$, the model behaves as a linear oscillator, not being able to exhibit the modulation presented before. Also, the model for $U_r = 8$ is not fully able to well represent the dynamics of the two distinct frequencies, as it seems only one is effectively present in the signal. This is verified by Figure 6, where only one frequency is recovered by the PSD, whereas two distinct are present in the original signal (Figure 2). Despite those deficiencies, we can see that the overall amplitude produced by the model captures well the actual one, showing that the model (although simple) can be used as first-order approximation of the dynamics.

We remark the analysis presented here was performed considering only the fully developed portion of the signal, ignoring the transient part. In the Appendix, we also provide some analysis for the complete signal, with some discussion about the regression and the model produced.

4. DISCUSSIONS AND FUTURE WORK

In this paper, we propose a methodology for the modeling of time-series data, based on Takens' embedding theorem, and apply it to a Vortex-Induced Vibration (VIV) configuration where only solid-information is available. By delaying the solid time-series, we manage to construct an embedding copying the original system's dynamics, which is then transformed in (numerical) time-derivatives of the signal itself. This new coordinate system is then used to write the system's dynamics by a regression procedure where a nonlinear (polynomial) model is obtained by fitting the data. This model reconstructs very well the dynamics provided a sufficient amount of delayed-coordinates are considered. Indeed, for re-

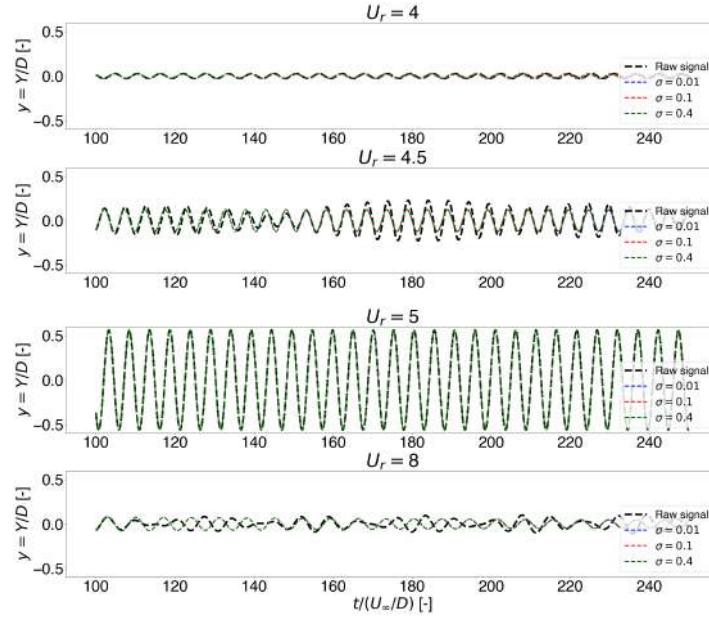


Figure 5. Signal from cross-wise displacement (y) for the model's outputs (in colors) and the comparison with original, raw signal, in black dashed lines, for $U_r = 4, 4.5, 5$ and 8

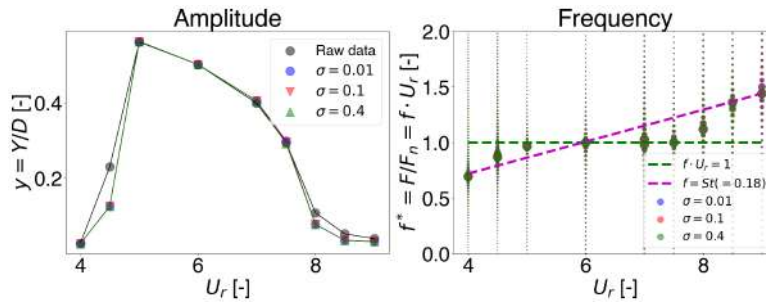


Figure 6. Signal from cross-wise displacement (y) and force (C_y) acting on the cylinder for $U_r = 4, 4.5, 5$ and 8

duced velocities completely outside or completely inside the lock-in, only one frequency is present in the signal, and tree delayed coordinates (rewritten as position, velocity and acceleration) are enough to close the dynamics of the attractor. For the cases $U_r = 4.5$ and $U_r = 8$, we obtain a transition signal, where two frequencies (the solid's natural frequency and the fluid's vortex-shedding one) coexist. In those cases, only sub-optimal results are obtained when considering three delayed coordinates, meaning that probably more should be added. The rewriting of those extra coordinates, either in terms of time-derivatives or delayed first and second ones is a direction of future work, as it may impact the quality of the model, together with its physical interpretability. Also, in this work we have considered a model where the inertia (expansion at the coefficients $-\gamma_{k,1}$) is free to vary. For this reason, a direction of future work would be on how to reinterpret this model in the light of the theory of systems with variable mass, as in Pesce (2003) (the added mass concept effects being intrinsically embedded in it).

5. ACKNOWLEDGEMENTS

The authors acknowledge the support by FAPESP, under the research grant number 2022/00770-0 and fellowship numbers 2024/03127-6 and 2024/02122-0. We also acknowledge CNPq support, under the grant numbers 306667/2023-1 and 307995/2022-4.

6. REFERENCES

- Adrian, R.J. and Westerweel, J., 2011. *Particle image velocimetry*. 30. Cambridge university press.
- Anderson, J.D. and Wendt, J., 1995. *Computational fluid dynamics*, Vol. 206. Springer.
- Assi, G.d.S., Bearman, P.W., Carmo, B.S., Meneghini, J.R., Sherwin, S.J. and Willden, R., 2013. "The role of wake stiffness on the wake-induced vibration of the downstream cylinder of a tandem pair". *Journal of Fluid Mechanics*, Vol. 718, pp. 210–245.

- Bearman, P.W., 1984. “Vortex shedding from oscillating bluff bodies”. *Annual review of fluid mechanics*, Vol. 16, pp. 195–222.
- Bhatti, M.M., Marin, M., Zeeshan, A. and Abdelsalam, S.I., 2020. “Recent trends in computational fluid dynamics”. *Frontiers in Physics*, Vol. 8, p. 593111.
- Bittnar, Z. and Sejnoha, J., 1996. *Numerical methods in structural mechanics*. Thomas Telford.
- Cunha, L.D., Pesce, C.P., Wanderley, J. *et al.*, 2006. “The robustness of the added mass in viv models”. In *Volume 4: Terry Jones Pipeline Technology; Ocean Space Utilization; CFD and VIV Symposium*. ASMEDC.
- Defensor Filho, W.A. and Pesce, C.P., 2023. “An experimental investigation with vertically suspended flexible pipes conveying water in aspirating and discharging conditions in a towing tank”.
- Franzini, G.R., Pereira, A.A., Fajarra, A.L. and Pesce, C.P., 2008. “Experiments on viv under frequency modulation and at constant reynolds numbers”. In *International Conference on Offshore Mechanics and Arctic Engineering*. Vol. 48227, pp. 959–968.
- Franzini, G.R., Fajarra, A.L.C., Meneghini, J.R., Korkischko, I. and Franciss, R., 2009. “Experimental investigation of vortex-induced vibration on rigid, smooth and inclined cylinders”. *Journal of Fluids and Structures*, Vol. 25, No. 4, pp. 742–750.
- Franzini, G.R., Gonçalves, R.T., Rosetti, G.F., Meneghini, J.R. and Fajarra, A.L.C., 2015. “Flow around circular cylinders with very low aspect ratio”. *Journal of Fluids and Structures*, Vol. 54, pp. 122–141.
- Fajarra, A.L.C., 2002. “Estudos experimentais e analíticos das vibrações induzidas pela emissão de vórtices em cilindros flexíveis e rígidos”. *São Paulo. Tese (Doutorado em Engenharia). USP*.
- Fajarra, A.L.C., Pesce, C.P., Flemming, F. and Williamson, C., 2001. “Vortex-induced vibration of a flexible cantilever”. *Journal of Fluids and Structures*, Vol. 15, No. 3-4, pp. 651–658.
- Hecht, F., 2012. “New development in freefem++”. *J. Numer. Math.*, Vol. 20, No. 3-4, pp. 251–265. ISSN 1570-2820. URL <https://freefem.org/>.
- Hirt, C., Amsden, A. and Cook, J., 1997. “An arbitrary lagrangian–eulerian computing method for all flow speeds”. *Journal of computational physics*, Vol. 135, No. 2, pp. 203–216.
- Hou, G., Wang, J. and Layton, A., 2012. “Numerical methods for fluid-structure interaction—a review”. *Communications in Computational Physics*, Vol. 12, No. 2, pp. 337–377.
- Kaiser, E., Kutz, J.N. and Brunton, S.L., 2018. “Sparse identification of nonlinear dynamics for model predictive control in the low-data limit”. *Proceedings of the Royal Society A*, Vol. 474, No. 2219, p. 20180335.
- Karniadakis, G.E., Israeli, M. and Orszag, S.A., 1991. “High-order splitting methods for the incompressible navier-stokes equations”. *Journal of computational physics*, Vol. 97, No. 2, pp. 414–443.
- Madi, L.S., Pesce, C.P. and Franzini, G.R., 2019. “Flexible piezoelectric cantilever under vortex-induced vibrations”. *DINAME 2019 - Proceedings of the XVIII International Symposium on Dynamic Problems of Mechanics*.
- Menon, K. and Mittal, R., 2021. “On the initiation and sustenance of flow-induced vibration of cylinders: insights from force partitioning”. *Journal of Fluid Mechanics*, Vol. 907, p. A37.
- Pesce, C.P. and Fajarra, A.L.C., 2000. “Vortex-induced vibrations and jump phenomenon: experiments with a clamped flexible cylinder in water”. *International Journal of Offshore and Polar Engineering*, Vol. 10, No. 01.
- Pesce, C., 2003. “The application of lagrange equations to mechanical systems with mass explicitly dependent on position”. *J. Appl. Mech.*, Vol. 70, No. 5, pp. 751–756.
- Sarpkaya, T., 1979. “Vortex-induced oscillations: a selective review”.
- Sauer, T., Yorke, J.A. and Casdagli, M., 1991. “Embedology”. *Journal of statistical Physics*, Vol. 65, pp. 579–616.
- Takens, F., 2006. “Detecting strange attractors in turbulence”. In *Dynamical Systems and Turbulence, Warwick 1980: proceedings of a symposium held at the University of Warwick 1979/80*. Springer, pp. 366–381.

7. RESPONSIBILITY NOTICE

The authors are solely responsible for the printed material included in this paper.

8. APPENDIX: MODEL WITH TRANSIENT

In this section, we briefly discuss on the use of the transient behavior to produce a model that also captures the dynamics outside the attractor. First, we should notice that the transient behavior, differently from the attractor, depends on the initial condition. Each different condition would produce a different path in the state-space $\mathbf{x} \in \mathbf{R}^n$, which can only be entirely “felt” by the delayed coordinates after a certain number of time-delays d which is certainly greater than the number needed for capturing the attractor only. If only a lower-dimension manifold is needed for representing an approximation of the transient dynamics, we can still hope to be economic on the embedding dimension d . In the following, we will present results taking the signal already presented in the body of this text, considering now the transient portion, which had as initial condition the fixed point (base-flow) to which the unstable vortex-shedding mode was added (multiplied by some small amplitude). This procedure helps to guide the modeling procedure to only represent the

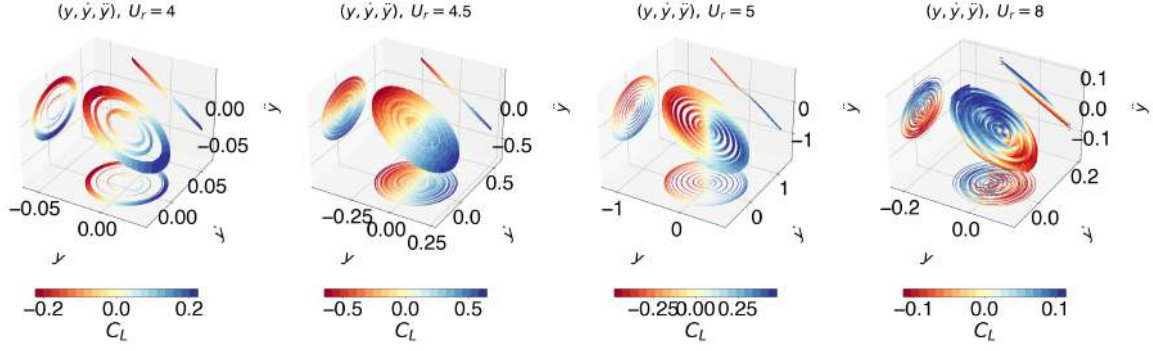


Figure 7. Same as Figure 3, but for the signal with transient part $t \geq 0$.

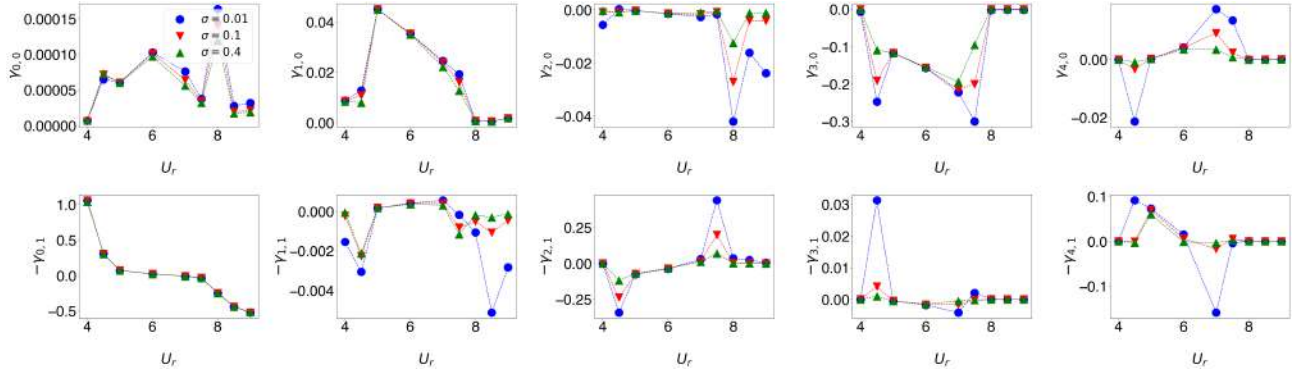


Figure 8. Same as Figure 4, but for the signal with transient part $t \geq 0$.

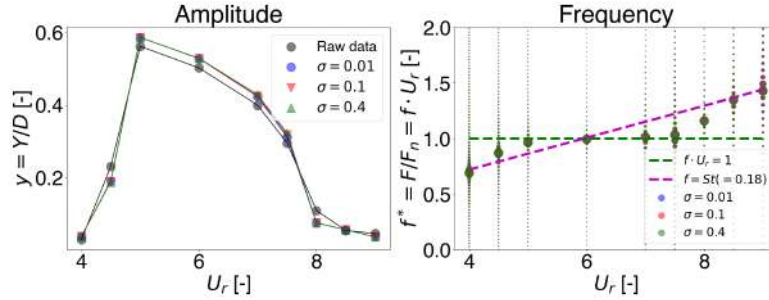


Figure 9. Same as Figure 6, but for the signal with transient part $t \geq 0$.

system's behavior on that unstable manifold, hoping that only the same three coordinates should be enough. For a more robust procedure, several initial conditions should be considered in the regression.

In Figure 7, we present again the trajectories on the $(y, \dot{y}, \ddot{y})$ space, now with the transient. Essentially, the main difference between Figure 7 and 3 is the extra information contained in the former. Indeed, the trajectories for $U_r = 4$ and 5 seem to lie on a disk rather than a circle, as previously shown and the disc for the case $U_r = 4.5$ seems to be "denser". Also, we can now appreciate better the fact that the lift C_L is indeed not a proper function of the triplet $(y, \dot{y}, \ddot{y})$.

Let us present now the model regression results. The coefficients are given in Figure 8 and a lock-in curve in Figure 9. We can see that the coefficients have similar behavior as before: the added mass effect and the linear and cubic damping are robustly obtained and other higher-order terms are shrunk to zero as more regularization is considered. Interestingly, the amplitude, as function of U_r , seems to be now over-predicted. This means that, to capture the transient behavior, the regressed model needs to be less "stable", leading to a nonlinear attractor with higher amplitude.